

Hacking Non-GPS/GPS-denied Drones

Security Analysis of LiDAR-based UAV Navigation and MAVLink Communication Protocols

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1. Introduction

Unmanned Aerial Vehicles (UAVs) increasingly rely on non-GPS navigation in environments where satellite signals are unavailable or unreliable. In this project, a **Holybro X500 running PX4** uses LiDAR-based localisation instead of GPS. While this improves navigation in indoor or GPS-denied settings, it also introduces new attack surfaces at both the sensor layer and the communication/protocol layer.

This research investigates how an attacker could disrupt UAV behaviour not necessarily by fully hijacking the drone, but by **corrupting navigation inputs, causing estimator instability, or triggering unsafe autonomy and failsafe decisions**.

2. Research Motivation

Two major security concerns were identified:

LiDAR Sensor Attacks

LiDAR can potentially be disrupted through spoofing, jamming, or environmental deception. These attacks may cause false range readings, dropped measurements, scan-matching failure, or localisation drift.

MAVLink Timing Vulnerabilities

A reviewed paper on *Timestamp Manipulation-Based GPS Spoofing Attacks on the MAVLink 2.0 Protocol* showed that MAVLink 2.0 signing can be undermined under certain conditions by manipulating time synchronization. The attack does not directly grant full control, but enables replay, denial of service, and persistent disruption.

3. System Architecture

The target system consists of:

- Holybro X500 UAV
- PX4 autopilot
- LiDAR-based navigation
- External odometry/pose estimation (companion computer)
- MAVLink communication with Ground Control Station (GCS)

Security Layer Architecture

Communication Layer: MAVLink Timing & Authentication

Control Layer: PX4 Estimator & Failsafe Logic

Perception Layer: LiDAR & State Estimation

4. Key Research Questions

- How can LiDAR-based navigation be disrupted in a GPS-denied PX4 UAV?
- What are the safety impacts of spoofed, delayed, stale, or dropped LiDAR-derived measurements?
- Can timing inconsistencies in MAVLink or external odometry create estimator failures or dangerous control decisions?
- Which countermeasures are most effective for improving resilience?

5. MAVLink Protocol Findings

The reviewed MAVLink paper demonstrated that:

- MAVLink 2.0 message signing depends on a **monotonically increasing timestamp**
- If time synchronization is influenced by spoofed GPS input, attackers may push timestamps forward

This Can Cause:

- Replay attacks under limited conditions
- Denial of service
- Timestamp overflow / persistent communication disruption
- However:** Cannot provide arbitrary command injection or total drone takeover without the signing key

Reframed Threat Model: The main danger is loss of trusted communication and safe autonomy, not necessarily classic "remote hijacking".

6. Theoretical Attack Pathways

1. LiDAR Spoofing / False Returns

Injected or misleading optical signals may cause incorrect distance reporting, affecting altitude or obstacle perception.

2. LiDAR Jamming / Measurement Dropout

Overloading the sensor or creating interference may cause range updates to disappear or become unreliable.

3. Environmental Deception

Reflective surfaces, repetitive geometry, or adversarial scene structure may confuse LiDAR odometry/SLAM.

4. Latency and Stale Odometry

Heavy computation or disrupted sensing can delay external pose updates, creating timestamp inconsistencies.

5. MAVLink Timing Manipulation

Protocol-level timing attacks may interfere with trusted communication streams.

7. Security Impact on PX4

In a LiDAR-based PX4 system, successful attacks may not directly "take over" the drone, but can produce serious outcomes:

- Position drift
- Localisation loss
- Unsafe mode switching
- Failsafe triggers
- Rejected external odometry
- Oscillatory or unstable behaviour
- Mission failure

Critical Security Goals: Availability, timing integrity, and estimator trust.

8. Experimental Direction

The planned controlled evaluation focuses on:

- Bench or lab-based testing on owned hardware
- Monitoring:** distance_sensor, vehicle_local_position, estimator_status
- Observing:** innovation spikes, covariance growth, sensor dropouts, odometry rejection, state transitions and failsafe behaviour

The goal is to **measure how navigation degrades under attack-like conditions** rather than to demonstrate offensive takeover.

9. Proposed Defences

Bound timestamp jumps – Prevent large temporal discontinuities

Decouple signing time from GNSS – Use independent time source

Stronger key management – Enhanced cryptographic practices

Odometry integrity checks – Validate external vision/odometry messages

MAVLink gateway validation – Add timing/quality checks

Multi-sensor cross-checking – Implement graceful degradation

10. Research Contribution

This dissertation aims to contribute a **combined security and safety analysis** of LiDAR-based UAV navigation under both:

- Sensor-layer disruption
- Protocol-layer timing manipulation

The main contribution is not proving full hijack, but showing how **subtle integrity and timing failures can cascade into unsafe autonomy** in PX4-based UAVs.

11. Conclusion

LiDAR-based, GPS-denied UAV navigation improves operational flexibility but introduces new vulnerabilities. Both sensor deception and MAVLink timing weaknesses can undermine navigation reliability, estimator trust, and safe autonomous behaviour.

Strengthening UAV resilience therefore requires protection across both the physical sensing layer and the communication layer.